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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 26th July Morning Shift

JEE Main Question Paper with Answer Key

26 Jul 2022	90	18	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	24 questions
Mathematics	21 questions
Physics	27 questions
Out of syllabus	18 questions
Answer key	at the end

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Chemistry 24 questions

Q1. MCQ (Single Correct Answer)

Open solution

Match List-I with List-II :

	List I (Compound)		List II (Shape)
(A)	BrF_5	(I)	bent
(B)	$[CrF_6]^{3-}$	(II)	square pyramidal
(C)	O_3	(III)	trigonal bipyramidal
(D)	PCl_5	(IV)	octahedral

Choose the correct answer from the options given below :

(A) $(A)-(I),(B)-(II),(C)-(III),(D)-(IV)$

(B) $(A)-(IV),(B)-(III),(C)-(II),(D)-(I)$

(C) $(A)-(II),(B)-(IV),(C)-(I),(D)-(III)$

(D) $(A)-(III),(B)-(IV),(C)-(II),(D)-(I)$

Q2. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

	List I (Processes / Reactions)		List II (Catalyst)
(A)	$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	(I)	Fe(s)
(B)	$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$	(II)	Pt(s) – Rh(s)
(C)	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(III)	V_2O_5
(D)	Vegetable oil(l) + $H_2 \rightarrow$ Vegetable ghee(s)	(IV)	Ni(s)

Choose the correct answer from the options given below :

(A) $(A)-(III),(B)-(I),(C)-(II),(D)-(IV)$

(B) $(A)-(III),(B)-(II),(C)-(I),(D)-(IV)$

(C) $(A)-(IV),(B)-(III),(C)-(I),(D)-(II)$

(D) $(A)-(IV),(B)-(II),(C)-(III),(D)-(I)$

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Q3. MCQ (Single Correct Answer)

Given two statements below :

Statement I : In Cl_2 molecule the covalent radius is double of the atomic radius of chlorine.

Statement II : Radius of anionic species is always greater than their parent atomic radius.

Choose the most appropriate answer from options given below :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Open solution

Q4. MCQ (Single Correct Answer)

Which of the given reactions is not an example of disproportionation reaction?

(A) $2H_2O_2 \rightarrow 2H_2O + O_2$

(B) $2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

(C) $MnO_4^- + 4H^+ + 3e^- \rightarrow MnO_2 + 2H_2O$

(D) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

Open solution

Q5. MCQ (Single Correct Answer)

The dark purple colour of $KMnO_4$ disappears in the titration with oxalic acid in acidic medium. The overall change in the oxidation number of manganese in the reaction is :

(A) 5 (B) 1 (C) 7 (D) 2

Open solution

Q6. MCQ (Single Correct Answer)

$\dot{Cl} + CH_4 \rightarrow A + B$

A and B in the above atmospheric reaction step are :

(A) C_2H_6 and Cl_2

(B) $\dot{C}HCl_2$ and H_2

(C) $\dot{C}H_3$ and HCl

(D) C_2H_6 and HCl

Open solution

Q7. MCQ (Single Correct Answer)

Which technique among the following, is most appropriate in separation of a mixture of $100\ mg$ of p-nitrophenol and picric acid ?

(A) Steam distillation

(B) 2-5 ft long column of silica gel

(C) Sublimation

(D) Preparative TLC (Thin Layer Chromatography)

Open solution

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Q8. MCQ (Single Correct Answer)

The difference in the reaction of phenol with bromine in chloroform and bromine in water medium is due to :

(A) Hyperconjugation in substrate

(B) Polarity of solvent

(C) Free radical formation

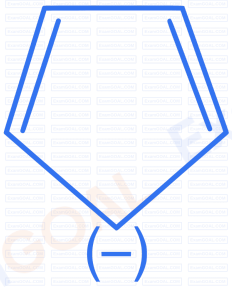
(D) Electromeric effect the substrate

Open solution

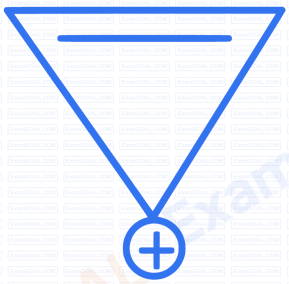
Q9. MCQ (Single Correct Answer)

Which of the following compounds is not aromatic?

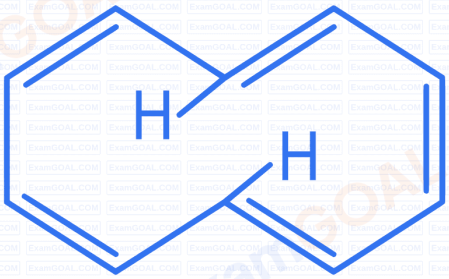
(A)



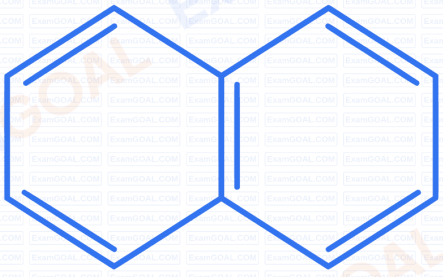
(B)



(C)



(D)



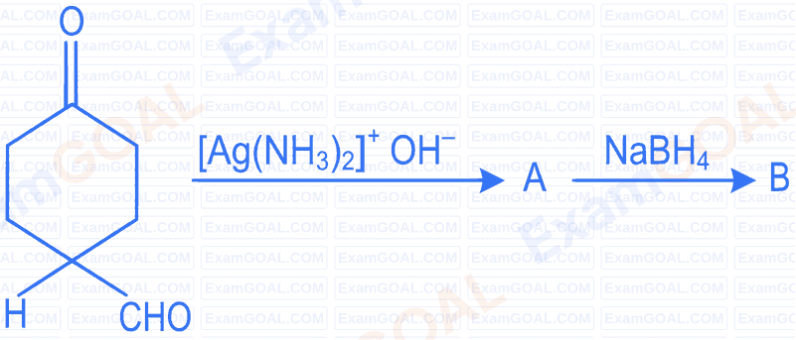
Open solution

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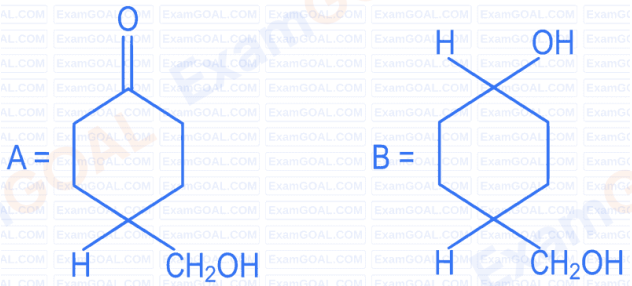
Q10. MCQ (Single Correct Answer)

Open solution

The products formed in the following reaction, A and B are



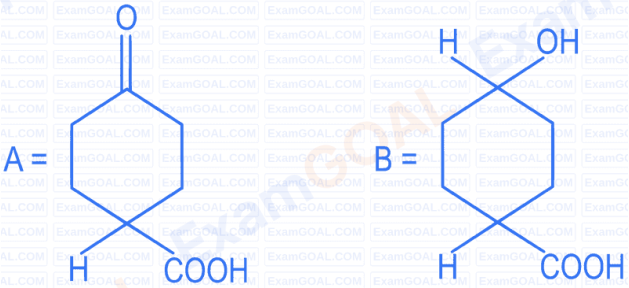
(A)



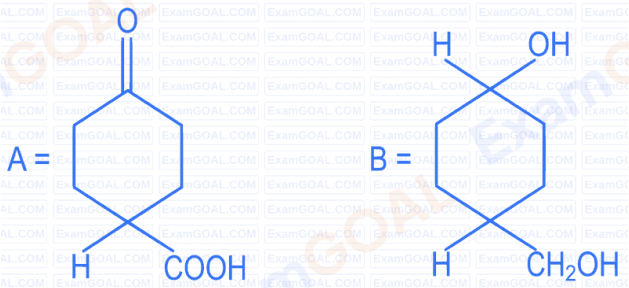
(B)



(C)



(D)

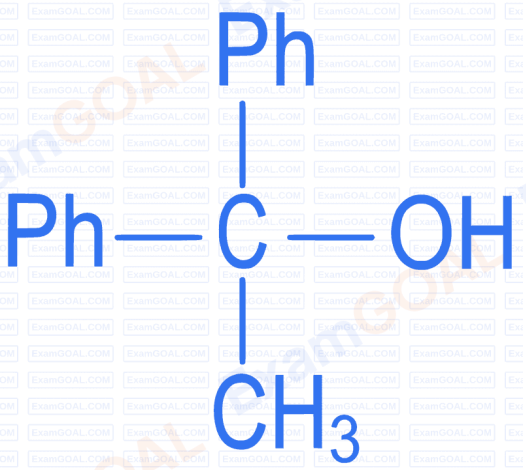


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Q11. MCQ (Single Correct Answer)

Open solution

Which reactant will give the following alcohol on reaction with one mole of phenyl magnesium bromide (PhMgBr) followed by acidic hydrolysis?



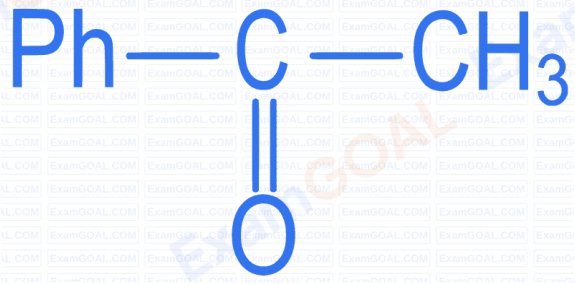
(A) $CH_3 - C \equiv N$

(B) $Ph - C \equiv N$

(C)



(D)

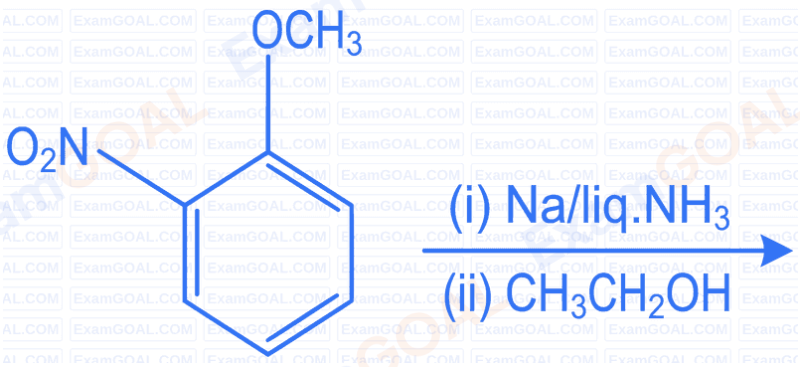


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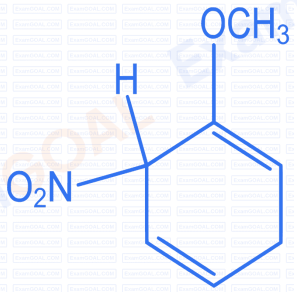
Q12. MCQ (Single Correct Answer)

Open solution

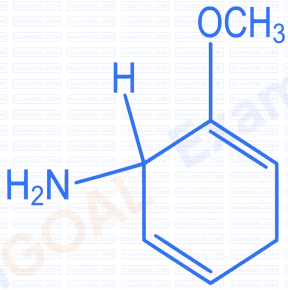
The major product of the following reaction is



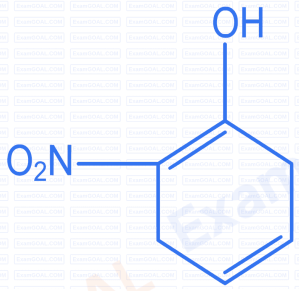
(A)



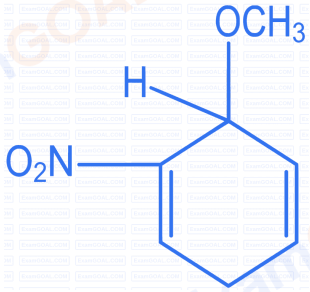
(B)



(C)



(D)

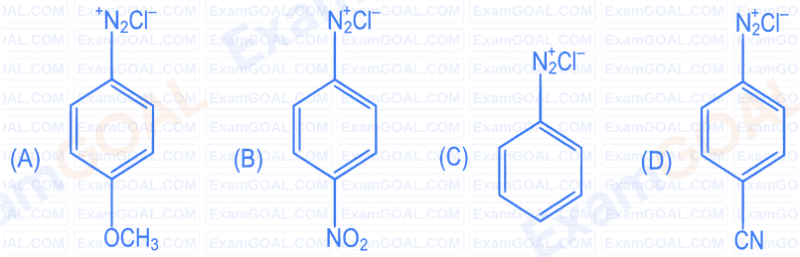


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Q13. MCQ (Single Correct Answer)

Open solution

The correct stability order of the following diazonium salt is



(A) $(A) > (B) > (C) > (D)$

(B) $(A) > (C) > (D) > (B)$

(C) $(C) > (A) > (D) > (B)$

(D) $(C) > (D) > (B) > (A)$

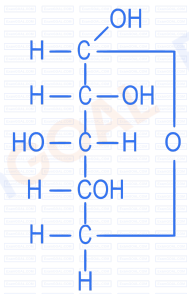
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Q14. MCQ (Single Correct Answer)

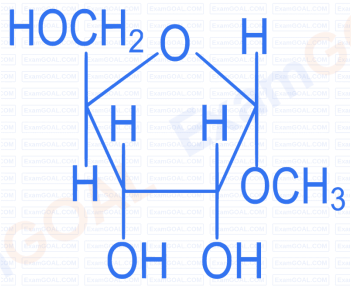
Open solution

Which one of the following is a reducing sugar?

(A)



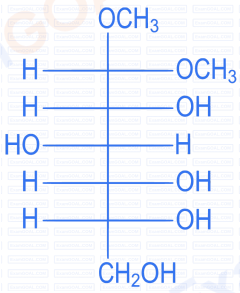
(B)



(C)



(D)



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Q15. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Experimental reaction of CH_3Cl with aniline and anhydrous $AlCl_3$ does not give o and p-methylaniline.

Reason (R): The $-NH_2$ group of aniline becomes deactivating because of salt formation with anhydrous $AlCl_3$ and hence yields m-methyl aniline as the product.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are true and (R) is the correct explanation of (A).

(B) Both A and R are true but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Q16. Numerical

Open solution

Chlorophyll extracted from the crushed green leaves was dissolved in water to make $2\ L$ solution of Mg of concentration $48\ ppm$. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of x is _____. (Nearest Integer)

(Given : Atomic mass of Mg is $24\ g\ mol^{-1}$; $N_A = 6.02 \times 10^{23}\ mol^{-1}$)

Q17. Numerical

Open solution

The wavelength of an electron and a neutron will become equal when the velocity of the electron is x times the velocity of neutron. The value of x is _____. (Nearest Integer)

(Mass of electron is $9.1 \times 10^{-31}\ kg$ and mass of neutron is $1.6 \times 10^{-27}\ kg$)

Q18. Numerical

Open solution

$2.4\ g$ coal is burnt in a bomb calorimeter in excess of oxygen at $298\ K$ and $1\ atm$ pressure. The temperature of the calorimeter rises from $298\ K$ to $300\ K$. The enthalpy change during the combustion of coal is $-x\ kJ\ mol^{-1}$. The value of x is _____. (Nearest Integer)

(Given : Heat capacity of bomb calorimeter $20.0\ kJ\ K^{-1}$. Assume coal to be pure carbon)

Q19. Numerical

Open solution

When 800 mL of 0.5 M nitric acid is heated in a beaker, its volume is reduced to half and 11.5 g of nitric acid is evaporated. The molarity of the remaining nitric acid solution is x $\times 10^{-2}$ M. (Nearest integer)

(Molar mass of nitric acid is 63 g mol^{-1})

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Q20. Numerical

At $298\ \mathrm{K}$, the equilibrium constant is 2×10^{15} for the reaction :

$\mathrm{Cu(s) + 2Ag^{+}(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)}$

The equilibrium constant for the reaction

$\mathrm{\frac{1}{2}Cu^{2+}(aq) + Ag(s) \rightleftharpoons \frac{1}{2}Cu(s) + Ag^{+}(aq)}$

is $x \times 10^{-8}$. The value of x is _____. (Nearest Integer)

Open solution

Q21. Numerical

The amount of charge in F (Faraday) required to obtain one mole of iron from $\mathrm{Fe_3O_4}$ is _____. (Nearest Integer)

Open solution

Q22. Numerical

For a reaction $\mathrm{A \rightarrow 2\,B + C}$ the half lives are $100\ \mathrm{s}$ and $50\ \mathrm{s}$ when the concentration of reactant A is 0.5 and $1.0\ \mathrm{mol\,L^{-1}}$ respectively. The order of the reaction is _____ . (Nearest Integer)

Open solution

Q23. Numerical

The difference between spin only magnetic moment values of $\mathrm{[Co(H_2O)_6]Cl_2}$ and $\mathrm{[Cr(H_2O)_6]Cl_3}$ is _____.

Open solution

Q24. Numerical

In the presence of sunlight, benzene reacts with $\mathrm{Cl_2}$ to give product, X. The number of hydrogens in X is _____.

Open solution

Mathematics 21 questions

Q25. MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be a continuous function such that $f(3x) - f(x) = x$. If $f(8) = 7$, then $f(14)$ is equal to :

(A) 4 (B) 10 (C) 11 (D) 16

Open solution

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Q26. MCQ (Single Correct Answer)

Let O be the origin and A be the point $z_1 = 1 + 2i$. If B is the point z_2, $\mathrm{Re}(z_2) < 0$, such that OAB is a right angled isosceles triangle with OB as hypotenuse, then which of the following is NOT true?

(A) $\arg z_2 = \pi - \tan^{-1}3$

(B) $\arg(z_1 - 2z_2) = -\tan^{-1}\frac{4}{3}$

(C) $|z_2| = \sqrt{10}$

(D) $|2z_1 - z_2| = 5$

Open solution

Q27. MCQ (Single Correct Answer)

If the system of linear equations.

$8x + y + 4z = -2$

$x + y + z = 0$

$\lambda x - 3y = \mu$

has infinitely many solutions, then the distance of the point $\left(\lambda, \mu, -\frac{1}{2}\right)$ from the plane $8x + y + 4z + 2 = 0$ is :

(A) $3\sqrt{5}$

(B) 4

(C) $\frac{26}{9}$

(D) $\frac{10}{3}$

Open solution

Q28. MCQ (Single Correct Answer)

The odd natural number a, such that the area of the region bounded by y = 1, y = 3, x = 0, x = y^a is $\frac{364}{3}$, is equal to :

(A) 3

(B) 5

(C) 7

(D) 9

Open solution

Q29. MCQ (Single Correct Answer)

Consider two G.Ps. 2, 2^2, 2^3, ... and 4, 4^2, 4^3, ... of 60 and n terms respectively. If the geometric mean of all the 60 + n terms is $(2)^{\frac{225}{8}}$, then $\sum_{k=1}^{n} k(n-k)$ is equal to :

(A) 560

(B) 1540

(C) 1330

(D) 2600

Open solution

Q30. MCQ (Single Correct Answer)

If the function $f(x) = \begin{cases} \frac{\log_e(1-x+x^2)+\log_e(1+x+x^2)}{\sec x - \cos x} & , \quad x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right) - \{0\} \\ k & , \quad x = 0 \end{cases}$ is continuous at x = 0, then k is equal to:

(A) 1

(B) -1

(C) e

(D) 0

Open solution

Q31. MCQ (Single Correct Answer)

If $f(x) = \begin{cases} x + a & , \quad x \le 0 \\ |x - 4| & , \quad x > 0 \end{cases}$ and $g(x) = \begin{cases} x + 1 & , \quad x < 0 \\ (x-4)^2 + b & , \quad x \ge 0 \end{cases}$ are continuous on R, then $(gof)(2) + (fog)(-2)$ is equal to :

(A) -10

(B) 10

(C) 8

(D) -8

Open solution

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Q32. MCQ (Single Correct Answer)

Let $f(x) = \begin{cases} x^3 - x^2 + 10x - 7, & x \le 1 \\ -2x + \log_2(b^2 - 4), & x > 1 \end{cases}$.

Then the set of all values of b, for which f(x) has maximum value at x = 1, is :

(A) $(-6, -2)$ (B) $(2, 6)$

(C) $[-6, -2) \cup (2, 6]$ (D) $\left[-\sqrt{6}, -2\right) \cup \left(2, \sqrt{6}\right]$

Open solution

Q33. MCQ (Single Correct Answer)

If $\frac{dy}{dx} + 2y \tan x = \sin x,\ 0 < x < \frac{\pi}{2}$ and $y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of $y(x)$ is :

(A) $\frac{1}{8}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$

Open solution

Q34. MCQ (Single Correct Answer)

A point P moves so that the sum of squares of its distances from the points $(1, 2)$ and $(-2, 1)$ is 14. Let $f(x, y) = 0$ be the locus of P, which intersects the x-axis at the points A, B and the y-axis at the points C, D. Then the area of the quadrilateral ACBD is equal to :

(A) $\frac{9}{2}$ (B) $\frac{3\sqrt{17}}{2}$ (C) $\frac{3\sqrt{17}}{4}$ (D) 9

Open solution

Q35. MCQ (Single Correct Answer)

The length of the perpendicular from the point $(1, -2, 5)$ on the line passing through $(1, 2, 4)$ and parallel to the line $x + y - z = 0 = x - 2y + 3z - 5$ is :

(A) $\sqrt{\frac{21}{2}}$ (B) $\sqrt{\frac{9}{2}}$ (C) $\sqrt{\frac{73}{2}}$ (D) 1

Open solution

Q36. MCQ (Single Correct Answer)

Let $\vec{a} = \alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}, \alpha > 0$. If the projection of $\vec{a} \times \vec{b}$ on the vector $-\hat{i} + 2\hat{j} - 2\hat{k}$ is 30, then α is equal to :

(A) $\frac{15}{2}$ (B) 8 (C) $\frac{13}{2}$ (D) 7

Open solution

Q37. MCQ (Single Correct Answer)

Let E_1, E_2, E_3 be three mutually exclusive events such that $P(E_1) = \frac{2+3p}{6}, P(E_2) = \frac{2-p}{8}$ and $P(E_3) = \frac{1-p}{2}$. If the maximum and minimum values of p are p_1 and p_2, then $(p_1 + p_2)$ is equal to :

(A) $\frac{2}{3}$ (B) $\frac{5}{3}$ (C) $\frac{5}{4}$ (D) 1

Open solution

Q38. MCQ (Single Correct Answer)

$\tan\left(2\tan^{-1}\frac{1}{5} + \sec^{-1}\frac{\sqrt{5}}{2} + 2\tan^{-1}\frac{1}{8}\right)$ is equal to :

(A) 1 (B) 2 (C) $\frac{1}{4}$ (D) $\frac{5}{4}$

Open solution

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Q39. Numerical

If for some $\mathrm{p}, \mathrm{q}, \mathrm{r} \in \mathbf{R}$, not all have same sign, one of the roots of the equation $\left(\mathrm{p}^2+\mathrm{q}^2\right) x^2-2 \mathrm{q}(\mathrm{p}+\mathrm{r}) x+\mathrm{q}^2+\mathrm{r}^2=0$ is also a root of the equation $x^2+2 x-8=0$, then $\frac{\mathrm{q}^2+\mathrm{r}^2}{\mathrm{p}^2}$ is equal to _____,

Open solution

Q40. Numerical

The number of 5-digit natural numbers, such that the product of their digits is 36 , is _____.

Open solution

Q41. Numerical

The number of distinct real roots of the equation $x^5\left(x^3-x^2-x+1\right)+x\left(3 x^3-4 x^2-2 x+4\right)-1=0$ is _____.

Open solution

Q42. Numerical

If $\mathrm{n}(2 \mathrm{n}+1) \int_0^1(1-x^{\mathrm{n}})^{2 \mathrm{n}} \mathrm{d} x=1177 \int_0^1(1-x^{\mathrm{n}})^{2 \mathrm{n}+1} \mathrm{~d} x$, then $\mathrm{n} \in \mathbf{N}$ is equal to _____.

Open solution

Q43. Numerical

The equations of the sides AB, BC and CA of a triangle ABC are $2 x+y=0, x+\mathrm{p} y=15 \mathrm{a}$ and $x-y=3$ respectively. If its orthocentre is $(2, a),-\frac{1}{2}<\mathrm{a}<2$, then p is equal to _____.

Open solution

Q44. Numerical

Let the function $f(x)=2 x^2-\log_{\mathrm{e}} x, x>0$, be decreasing in $(0, \mathrm{a})$ and increasing in $(\mathrm{a}, 4)$. A tangent to the parabola $y^2=4 a x$ at a point P on it passes through the point $(8 \mathrm{a}, 8 \mathrm{a}-1)$ but does not pass through the point $\left(-\frac{1}{a}, 0\right)$. If the equation of the normal at P is : $\frac{x}{\alpha}+\frac{y}{\beta}=1$, then $\alpha+\beta$ is equal to _____.

Open solution

Q45. Numerical

Let Q and R be two points on the line $\frac{x+1}{2}=\frac{y+2}{3}=\frac{z-1}{2}$ at a distance $\sqrt{26}$ from the point $P(4,2,7)$. Then the square of the area of the triangle PQR is _____.

Open solution

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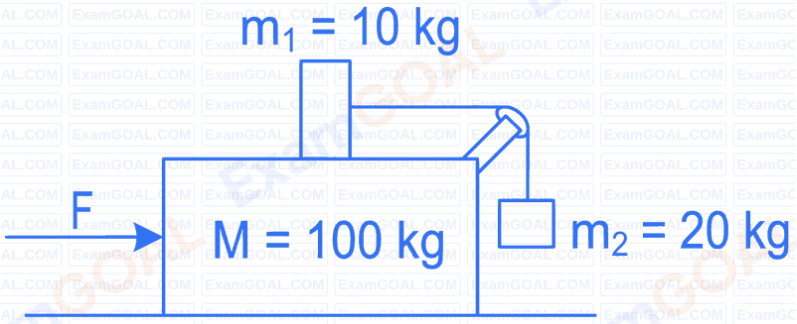
Physics 27 questions

Q46. MCQ (Single Correct Answer)

Open solution

Three masses $M = 100\,\text{kg}$, $m_1 = 10\,\text{kg}$ and $m_2 = 20\,\text{kg}$ are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force F is applied on the system so that the mass m_2 moves upward with an acceleration of $2\,\text{ms}^{-2}$. The value of F is :

(Take $g = 10\,\text{ms}^{-2}$)



(A) 3360 N (B) 3380 N (C) 3120 N (D) 3240 N

Q47. MCQ (Single Correct Answer)

Open solution

A parallel beam of light of wavelength $900\,\text{nm}$ and intensity $100\,\text{Wm}^{-2}$ is incident on a surface perpendicular to the beam. The number of photons crossing $1\,\text{cm}^2$ area perpendicular to the beam in one second is :

(A) 3×10^{16} (B) 4.5×10^{16} (C) 4.5×10^{17} (D) 4.5×10^{20}

Q48. MCQ (Single Correct Answer)

Open solution

In Young's double slit experiment, the fringe width is $12\,\text{mm}$. If the entire arrangement is placed in water of refractive index $\frac{4}{3}$, then the fringe width becomes (in mm):

(A) 16 (B) 9 (C) 48 (D) 12

Q49. MCQ (Single Correct Answer)

Open solution

The magnetic field of a plane electromagnetic wave is given by :

$\vec{B} = 2 \times 10^{-8} \sin\left(0.5 \times 10^3 x + 1.5 \times 10^{11} t\right) \hat{j}\,\text{T}$.

The amplitude of the electric field would be :

(A) $6\,\text{Vm}^{-1}$ along x-axis (B) $3\,\text{Vm}^{-1}$ along z-axis

(C) $6\,\text{Vm}^{-1}$ along z-axis (D) $2 \times 10^{-8}\,\text{Vm}^{-1}$ along z-axis

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Q50. MCQ (Single Correct Answer)

In a series LR circuit $X_L = R$ and power factor of the circuit is P_1. When capacitor with capacitance C such that $X_L = X_C$ is put in series, the power factor becomes P_2. The ratio $\frac{P_1}{P_2}$ is:

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{\sqrt{2}}$ (D) 2 : 1

Open solution

Q51. MCQ (Single Correct Answer)

A charge particle is moving in a uniform magnetic field $(2\hat{i} + 3\hat{j})\,\mathrm{T}$. If it has an acceleration of $(\alpha\hat{i} - 4\hat{j})\,\mathrm{m/s^2}$, then the value of α will be :

(A) 3 (B) 6 (C) 12 (D) 2

Open solution

Q52. MCQ (Single Correct Answer)

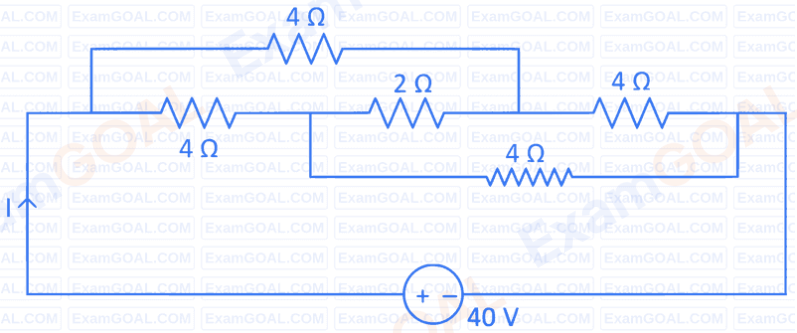
B_X and B_Y are the magnetic fields at the centre of two coils X and Y respectively each carrying equal current. If coil X has 200 turns and $20\ \mathrm{cm}$ radius and coil Y has 400 turns and $20\ \mathrm{cm}$ radius, the ratio of B_X and B_Y is :

(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 4 : 1

Open solution

Q53. MCQ (Single Correct Answer)

The current I in the given circuit will be :



(A) 10 A (B) 20 A (C) 4 A (D) 40 A

Open solution

Q54. MCQ (Single Correct Answer)

The total charge on the system of capacitors $C_1 = 1\mu\mathrm{F}, C_2 = 2\mu\mathrm{F}, \mathrm{C}_3 = 4\mu\mathrm{F}$ and $\mathrm{C}_4 = 3\mu\mathrm{F}$ connected in parallel is :

(Assume a battery of $20\ \mathrm{V}$ is connected to the combination)

(A) $200\ \mu\mathrm{C}$ (B) 200 C (C) $10\ \mu\mathrm{C}$ (D) 10 C

Open solution

Q55. MCQ (Single Correct Answer)

When a particle executes Simple Harmonic Motion, the nature of graph of velocity as a function of displacement will be :

(A) Circular (B) Elliptical (C) Sinusoidal (D) Straight line

Open solution

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Q56. MCQ (Single Correct Answer)

7 mol of a certain monoatomic ideal gas undergoes a temperature increase of $40\ \mathrm{K}$ at constant pressure. The increase in the internal energy of the gas in this process is :

(Given $\mathrm{R} = 8.3\ \mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$)

Open solution

(A) 5810 J (B) 3486 J (C) 11620 J (D) 6972 J

Q57. MCQ (Single Correct Answer)

A monoatomic gas at pressure P and volume V is suddenly compressed to one eighth of its original volume. The final pressure at constant entropy will be :

Open solution

(A) P (B) 8P (C) 32P (D) 64P

Q58. MCQ (Single Correct Answer)

A water drop of radius $1\ \mathrm{cm}$ is broken into 729 equal droplets. If surface tension of water is 75 dyne/ cm, then the gain in surface energy upto first decimal place will be :

(Given $\pi = 3.14$)

Open solution

(A) $8.5 \times 10^{-4}\ \mathrm{J}$ (B) $8.2 \times 10^{-4}\ \mathrm{J}$ (C) $7.5 \times 10^{-4}\ \mathrm{J}$ (D) $5.3 \times 10^{-4}\ \mathrm{J}$

Q59. MCQ (Single Correct Answer)

The percentage decrease in the weight of a rocket, when taken to a height of $32\ \mathrm{km}$ above the surface of earth will, be :

(Radius of earth $= 6400\ \mathrm{km}$)

Open solution

(A) 1% (B) 3% (C) 4% (D) 0.5%

Q60. MCQ (Single Correct Answer)

As per the given figure, two blocks each of mass $250\ \mathrm{g}$ are connected to a spring of spring constant $2\ \mathrm{Nm}^{-1}$. If both are given velocity v in opposite directions, then maximum elongation of the spring is :

Open solution



(A) $\frac{v}{2\sqrt{2}}$ (B) $\frac{v}{2}$ (C) $\frac{v}{4}$ (D) $\frac{v}{\sqrt{2}}$

Q61. MCQ (Single Correct Answer)

A monkey of mass $50\ \mathrm{kg}$ climbs on a rope which can withstand the tension (T) of $350\ \mathrm{N}$. If monkey initially climbs down with an acceleration of $4\ \mathrm{m/s^2}$ and then climbs up with an acceleration of $5\ \mathrm{m/s^2}$. Choose the correct option $\left(g = 10\ \mathrm{m/s^2}\right)$.

Open solution

(A) $T = 700\ \mathrm{N}$ while climbing upward (B) $T = 350\ \mathrm{N}$ while going downward

(C) Rope will break while climbing upward (D) Rope will break while going downward

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Q62. MCQ (Single Correct Answer)

Two projectiles thrown at 30° and 45° with the horizontal respectively, reach the maximum height in same time. The ratio of their initial velocities is :

(A) $1 : \sqrt{2}$ (B) $2 : 1$ (C) $\sqrt{2} : 1$ (D) $1 : 2$

Open solution

Q63. MCQ (Single Correct Answer)

A screw gauge of pitch $0.5\ \text{mm}$ is used to measure the diameter of uniform wire of length $6.8\ \text{cm}$, the main scale reading is $1.5\ \text{mm}$ and circular scale reading is 7 . The calculated curved surface area of wire to appropriate significant figures is :

[Screw gauge has 50 divisions on its circular scale]

(A) 6.8 cm^2 (B) 3.4 cm^2 (C) 3.9 cm^2 (D) 2.4 cm^2

Open solution

Q64. Numerical

If the initial velocity in horizontal direction of a projectile is unit vector $\hat{i}$ and the equation of trajectory is $y = 5x(1 - x)$. The y component vector of the initial velocity is _____ $\hat{j}$. $(\text{Take } \text{g} = 10\ \text{m/s}^2)$

Open solution

Q65. Numerical

A disc of mass $1\ \text{kg}$ and radius R is free to rotate about a horizontal axis passing through its centre and perpendicular to the plane of disc. A body of same mass as that of disc is fixed at the highest point of the disc. Now the system is released, when the body comes to the lowest position, its angular speed will be $4\sqrt{\frac{x}{3R}}\ \text{rad}\ s^{-1}$ where $x =$ _____. $\left(g = 10\ \text{ms}^{-2}\right)$

Open solution

Q66. Numerical

In an experiment to determine the Young's modulus of wire of a length exactly $1\ \text{m}$, the extension in the length of the wire is measured as $0.4\ \text{mm}$ with an uncertainty of $\pm 0.02\ \text{mm}$ when a load of $1\ \text{kg}$ is applied. The diameter of the wire is measured as $0.4\ \text{mm}$ with an uncertainty of $\pm 0.01\ \text{mm}$. The error in the measurement of Young's modulus (ΔY) is found to be $x \times 10^{10}\ \text{Nm}^{-2}$. The value of x is _____. (take $\text{g} = 10\ \text{ms}^{-2}$)

Open solution

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Q67. Numerical

A composite parallel plate capacitor is made up of two different dielectric materials with different thickness (t_1 and t_2) as shown in figure. The two different dielectric materials are separated by a conducting foil F. The voltage of the conducting foil is _____ V.

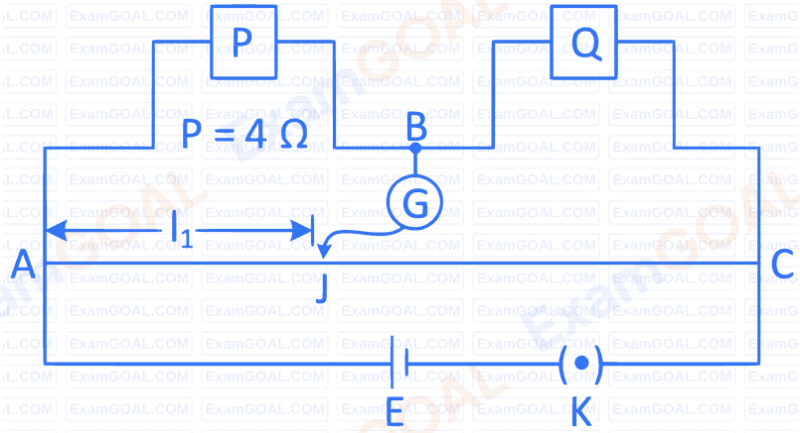
Open solution



Q68. Numerical

Resistances are connected in a meter bridge circuit as shown in the figure. The balancing length l_1 is 40 cm. Now an unknown resistance x is connected in series with P and new balancing length is found to be 80 cm measured from the same end. Then the value of x will be _____ Ω.

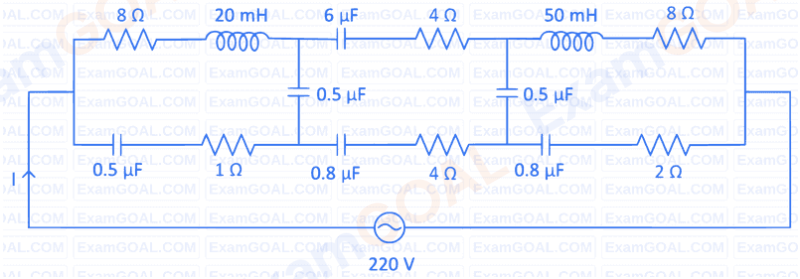
Open solution



Q69. Numerical

The effective current I in the given circuit at very high frequencies will be _____ A.

Open solution

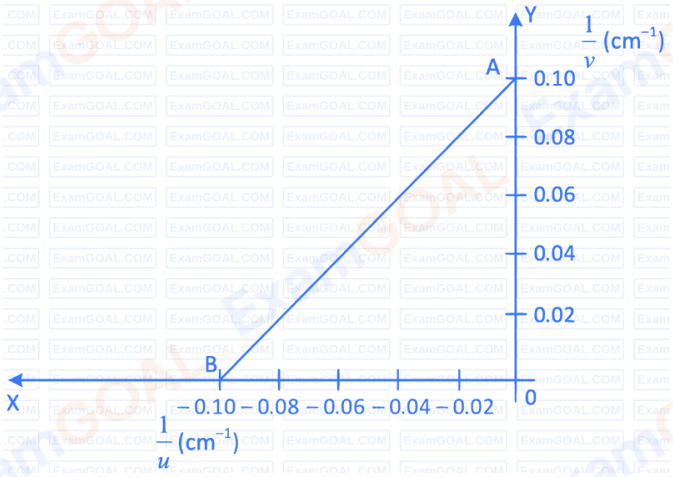


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Q70. Numerical

The graph between $\frac{1}{u}$ and $\frac{1}{v}$ for a thin convex lens in order to determine its focal length is plotted as shown in the figure. The refractive index of lens is 1.5 and its both the surfaces have same radius of curvature R. The value of R will be _____ cm. (where $u =$ object distance, $v =$ image distance)

Open solution



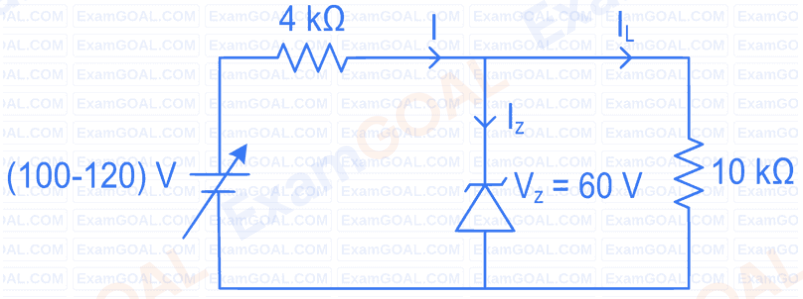
Q71. Numerical

In the hydrogen spectrum, λ be the wavelength of first transition line of Lyman series. The wavelength difference will be "aλ" between the wavelength of 3rd transition line of Paschen series and that of 2nd transition line of Balmer series where $\mathrm{a} =$ _____.

Open solution

Q72. Numerical

In the circuit shown below, maximum zener diode current will be _____ mA.



Open solution

Out of syllabus 18 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q73. MCQ (Single Correct Answer) · **Chemistry**

Refining using liquation method is the most suitable for metals with :

(A) Low melting point

(B) High boiling point

(C) High electrical conductivity

(D) Less tendency to be soluble in melts than impurities

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Which of the following can be used to prevent the decomposition of H_2O_2 ?

(A) Urea (B) Formaldehyde (C) Formic acid (D) Ethanol

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Reaction of $BeCl_2$ with $LiAlH_4$ gives :

(A) $AlCl_3$

(B) BeH_2

(C) LiH

(D) $LiCl$

(E) $BeAlH_4$

Choose the correct answer from options given below :

(A) (A), (D) and (E) (B) (A), (B) and (D) (C) (D) and (E) (D) (B), (C) and (D)

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Borazine, also known as inorganic benzene, can be prepared by the reaction of 3-equivalents of "X" with 6-equivalents of "Y". "X" and "Y", respectively are :

(A) $B(OH)_3$ and NH_3 (B) B_2H_6 and NH_3

(C) B_2H_6 and HN_3 (D) NH_3 and B_2O_3

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

Stearic acid and polyethylene glycol react to form which one of the following soap/s detergents ?

(A) Cationic detergent (B) Soap

(C) Anionic detergent (D) Non-ionic detergent

Open solution

Q78. Numerical · **Chemistry**

A mixture of hydrogen and oxygen contains 40% hydrogen by mass when the pressure is 2.2 bar. The partial pressure of hydrogen is _____ bar. (Nearest Integer)

Open solution

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Q79. MCQ (Single Correct Answer) · **Mathematics**

Let A be a 2 × 2 matrix with det (A) = − 1 and det ((A + I) (Adj (A) + I)) = 4. Then the sum of the diagonal elements of A can be :

(A) -1 (B) 2 (C) 1 (D) $-\sqrt{2}$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

If $a=\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\frac{2n}{n^2+k^2}$ and $f(x)=\sqrt{\frac{1-\cos x}{1+\cos x}}$, $x\in(0,1)$, then :

(A) $2\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$ (B) $f\left(\frac{a}{2}\right)f'\left(\frac{a}{2}\right)=\sqrt{2}$

(C) $\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$ (D) $f\left(\frac{a}{2}\right)=\sqrt{2}f'\left(\frac{a}{2}\right)$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let the tangent drawn to the parabola $y^2=24x$ at the point (α,β) is perpendicular to the line $2x+2y=5$. Then the normal to the hyperbola $\frac{x^2}{\alpha^2}-\frac{y^2}{\beta^2}=1$ at the point $(\alpha+4,\beta+4)$ does NOT pass through the point :

(A) (25, 10) (B) (20, 12) (C) (30, 8) (D) (15, 13)

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The mean and variance of a binomial distribution are α and $\frac{\alpha}{3}$ respectively. If $\mathrm{P}(X=1)=\frac{4}{243}$, then $\mathrm{P}(X=4 \text{ or } 5)$ is equal to :

(A) $\frac{5}{9}$ (B) $\frac{64}{81}$ (C) $\frac{16}{27}$ (D) $\frac{145}{243}$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let $S=\left\{\theta\in[0,2\pi]:8^{2\sin^2\theta}+8^{2\cos^2\theta}=16\right\}$. Then $n(S)+\sum\limits_{\theta\in S}\left(\sec\left(\frac{\pi}{4}+2\theta\right)\operatorname{cosec}\left(\frac{\pi}{4}+2\theta\right)\right)$ is equal to:

(A) 0 (B) -2 (C) -4 (D) 12

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

The statement $(\sim(\mathrm{p}\Leftrightarrow\sim\mathrm{q}))\wedge\mathrm{q}$ is :

(A) a tautology (B) a contradiction

(C) equivalent to $(p\Rightarrow q)\wedge q$ (D) equivalent to $(p\Rightarrow q)\wedge p$

Open solution

Q85. Numerical · **Mathematics**

The series of positive multiples of 3 is divided into sets : $\{3\},\{6,9,12\},\{15,18,21,24,27\},\ldots$ Then the sum of the elements in the 11th set is equal to _____.

Open solution

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Q86. Numerical · **Mathematics**

If the coefficients of x and x^2 in the expansion of $(1+x)^{\mathrm{p}}(1-x)^{\mathrm{q}}, \mathrm{p}, \mathrm{q} \leq 15$, are -3 and -5 respectively, then the coefficient of x^3 is equal to _____.

Open solution

Q87. Numerical · **Mathematics**

Let a curve $y = y(x)$ pass through the point $(3, 3)$ and the area of the region under this curve, above the x-axis and between the abscissae 3 and $x(> 3)$ be $\left(\frac{y}{x}\right)^3$. If this curve also passes through the point $(\alpha, 6\sqrt{10})$ in the first quadrant, then α is equal to _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

A radio can tune to any station in $6\ \mathrm{MHz}$ to $10\ \mathrm{MHz}$ band. The value of corresponding wavelength bandwidth will be :

(A) 4 m (B) 20 m (C) 30 m (D) 50 m

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

The disintegration rate of a certain radioactive sample at any instant is 4250 disintegrations per minute. 10 minutes later, the rate becomes 2250 disintegrations per minute. The approximate decay constant is :

(Take $\log_{10} 1.88 = 0.274$)

(A) $0.02\ \mathrm{min}^{-1}$ (B) $2.7\ \mathrm{min}^{-1}$ (C) $0.063\ \mathrm{min}^{-1}$ (D) $6.3\ \mathrm{min}^{-1}$

Open solution

Q90. Numerical · **Physics**

When a car is approaching the observer, the frequency of horn is $100\ \mathrm{Hz}$. After passing the observer, it is $50\ \mathrm{Hz}$. If the observer moves with the car, the frequency will be $\frac{x}{3}\ \mathrm{Hz}$ where $x =$ _____.

Open solution

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Answer Key

Chemistry

1. **C**	2. **B**	3. **D**	4. **C**	5. **A**
6. **C**	7. **D**	8. **B**	9. **C**	10. **C**
11. **D**	12. **A**	13. **B**	14. **A**	15. **C**
16. **24**	17. **1758**	18. **200**	19. **54**	20. **2**
21. **3**	22. **2**	23. **0**	24. **6**	

Mathematics

25. **B**	26. **D**	27. **D**	28. **B**	29. **C**
30. **A**	31. **D**	32. **C**	33. **A**	34. **B**
35. **A**	36. **D**	37. **B**	38. **B**	39. **272**
40. **180**	41. **3**	42. **24**	43. **3**	44. **45**
45. **153**				

Physics

46. **A**	47. **B**	48. **B**	49. **C**	50. **B**
51. **B**	52. **B**	53. **A**	54. **A**	55. **B**
56. **B**	57. **C**	58. **C**	59. **A**	60. **B**
61. **C**	62. **C**	63. **B**	64. **5**	65. **5**
66. **2**	67. **60**	68. **20**	69. **44**	70. **10**
71. **5**	72. **9**			

Out of syllabus

73. **A**	74. **A**	75. **B**	76. **B**	77. **D**
78. **2**	79. **B**	80. **C**	81. **D**	82. **C**
83. **C**	84. **D**	85. **6993**	86. **23**	87. **6**
88. **B**	89. **C**	90. **200**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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